

Statistical and Machine Learning

Linear models

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Linear models

- ▶ Linear regression
- ▶ Logistic regression
- ▶ Linear discriminant analysis

Linear models

- ▶ We only briefly review three basic linear models: linear regression, logistic regression and linear discriminant analysis (LDA).
- ▶ Most of the topics are covered in reference book [PML](#) Chapter 9-11 or [ESLII](#) Chapters 3.2, 4.3 and 4.4.
- ▶ You can also refer to other classes such as Multivariate Statistical Analysis, Statistical Inference I and II, or Generalized Regression.
- ▶ The purpose is to lay the foundations for follow-up techniques and to highlight connections between machine learning and statistics, rather than to go into the details of each method, as we assume that you have covered those in previous classes.

Linear models

- ▶ **Linear regression**
- ▶ Logistic regression
- ▶ Linear discriminant analysis

What is a linear model?

In a linear model, the response variable Y_i is assumed to be **linear in the parameters** $\beta_0, \beta_1, \dots, \beta_q$ and a random error term ε_i .

$$Y_i = \beta_0 + \sum_{j=1}^q \beta_j x_{ij} + \varepsilon_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i.$$

- ▶ Y_i : Dependent variables/response/target
- ▶ $\mathbf{x}_i$: Explanatory or independent variables/Predictors/Features
- ▶ β_0 : Regression parameters/Coefficients/Intercept+Slope.
- ▶ ε_i : Random error/Unexplained variation.
- ▶ It is often assumed that $\varepsilon_i \sim_{i.i.d.} (0, \sigma^2)$, i.e., $\mathbb{E}(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \mathbb{E}(\varepsilon_i^2) = \sigma^2$.
- ▶ These so-called Gauss-Markov conditions, can be expressed as: $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\mathbb{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T) = \sigma^2 \mathbf{I}_n$.

Linear regression in matrix form

We can re-write the linear model as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1q} \\ 1 & x_{21} & \cdots & x_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nq} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_q \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

Let $p = q + 1$. The $n \times p$ *design* or *model matrix* $\mathbf{X}$ satisfies:

- ▶ Each row represents a vector of covariates $\mathbf{x}_i = (1, x_{i1}, \dots, x_{iq})^\top$,
- ▶ These covariates are observed along with each response observation Y_i .

Notation and assumptions

- ▶ For the i -th subject, we observe $(y_i, \mathbf{x}_i)$, where $i = 1, \dots, n$, which refer to realizations of $Y_i, \mathbf{X}_i$.
- ▶ The j -th column of $\mathbf{X}$ contains all observations made on the j -th variable included in the model, for $j = 1, \dots, p$.
- ▶ Throughout this lecture we assume:
 - ▶ $n > p$,
 - ▶ $\mathbf{X}$ has full column rank, i.e., $\text{rank}(\mathbf{X}) = p$,
 - ▶ $\mathbf{X}^\top \mathbf{X}$ is invertible.

Notation and assumptions

- ▶ The response variable Y_i satisfies: $\mathbb{E}(Y_i) = \mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, $\text{var}(Y_i) = \sigma^2$ which is denoted as:

$$Y_i \sim (\mu_i, \sigma^2) \quad \text{or} \quad \mathbf{Y} \sim (\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$$

- ▶ For statistical inference purposes (testing hypotheses or finding confidence intervals for β_j) we often assume that the random error ε follows a normal distribution: $\varepsilon \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$
- ▶ Under this assumption, Y_i satisfies:

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2) \quad \text{or} \quad \mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$$

- ▶ While the observations $(y_i, \mathbf{x}_i)$ are often assumed to be *independent* of one another, the covariates are typically *correlated*, meaning $\mathbf{X}^\top \mathbf{X}$ is not assumed to be diagonal.

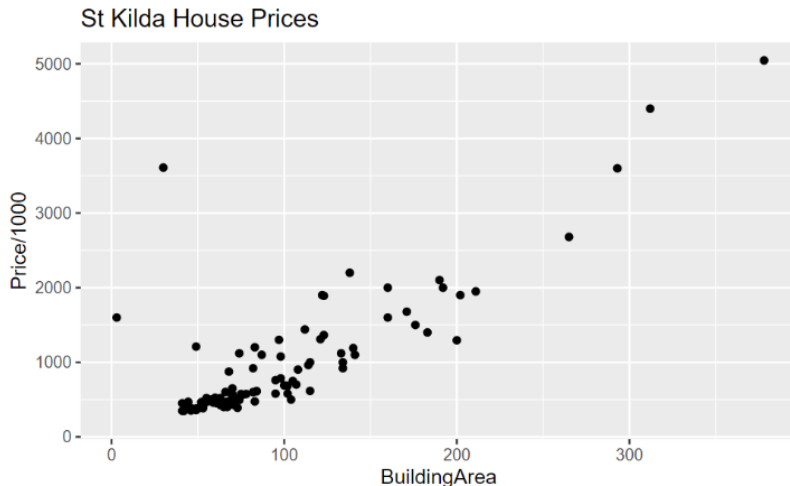
Covariates

Covariates / features / regressors are either *numeric* or *categorical*.

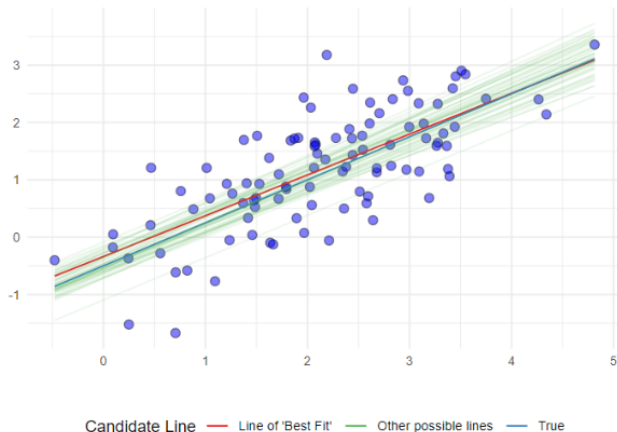
- ▶ Numeric variables take on meaningful numeric quantities. They are further classified into:
 - ▶ *Continuous variables* — can take values anywhere in $\mathbb{R}$
 - ▶ *Discrete variables* — can only take values from a countable set, e.g. $\mathbb{N}^+ = 0 \cup \mathbb{N} = \{0, 1, 2, \dots\}$
- ▶ Categorical variables, sometimes called *factors*, are not numeric and take on specified levels. They are further classified into:
 - ▶ *Ordered categorical variables* — their levels are ordered, e.g. house size $\in \{\text{small, medium, large}\}$
 - ▶ *Non-ordered categorical variables* — no clear ordering can be defined, e.g. house type $\in \{\text{bungalow, detached, semi-detached}\}$

Example – house price prediction

What is the price of a 100 sqm house in St Kilda?



Least squares estimation



- ▶ Data simulated from $y_i = -0.5 + 0.75x_i + \varepsilon_i$
- ▶ True line are shown in blue.
- ▶ Least square fit is shown in red.
- ▶ Why don't we choose one of the green lines?

Least square estimation

- ▶ An estimator of the vector of model parameters β_0 can be obtained by *the method of ordinary least squares* (OLS), which is one of the most widely used methods of estimation besides maximum likelihood estimation.
- ▶ The aim is to find some $\beta \in \mathbb{R}^p$ which minimises the “distance” between $\mathbf{Y}$ and $\mathbf{X}\beta$ by minimising the residual sum of squares (RSS) with respect to $\beta \in \mathbb{R}^p$:

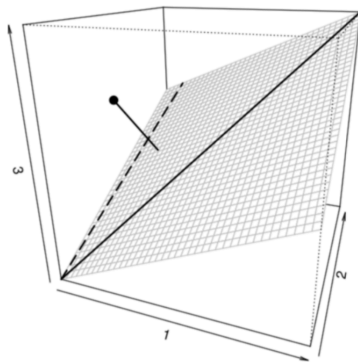
$$S(\beta) = \sum_{i=1}^n (Y_i - \mathbf{x}_i^\top \beta)^2 = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2$$

- ▶ By finding the *minimiser* of $S(\beta)$, say $\hat{\beta}$, we find *the linear combination of the column vectors* of $\mathbf{X}$, namely $\mathbf{X}\hat{\beta}$, which minimises the Euclidean distance between $\mathbf{Y}$ and $\mathbf{X}\beta$ among all possible $\beta \in \mathbb{R}^p$.

OLS – geometry

$$\mathbf{Y} = \begin{bmatrix} 0.05 \\ 0.40 \\ 0.65 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & 0.1 \\ 1 & 1 \\ 1 & 0.6 \end{bmatrix}$$

The OLS $\hat{\mu}$ of $\mu = \mathbb{E}(\mathbf{Y})$ is the orthogonal projection of the data point $\mathbf{Y}$ onto the subspace spanned by the columns of $\mathbf{X}$. $\hat{\mu}$ is the closest to $\mathbf{Y}$ according to the Euclidean distance. Taken from Wood, 2015.



The point represents $\mathbf{Y}$, while the solid and broken lines represent the vectors in the directions of the two columns of $\mathbf{X}$, and the subspace spanned by the two vectors is shaded in grey.

OLS – analytical solution

Solving $\hat{\beta} = \arg \min_{\beta} S(\beta)$ yields

$$\hat{\beta} = (\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{X}^{\top} \mathbf{Y}.$$

The assumption $\varepsilon \sim (\mathbf{0}, \sigma^2 \mathbf{I}_n)$ yields $\mathbf{Y} \sim (\mathbf{X} \beta_0, \sigma^2 \mathbf{I}_n)$.

We then have $\mathbb{E}(\hat{\beta}) = \beta_0$ and $\text{cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^{\top} \mathbf{X})^{-1}$.

$\hat{\beta}$ is a *linear* transformation of $\mathbf{Y}$. Hence, under the additional assumption of normality $\varepsilon \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, we have

$$\hat{\beta} \sim \mathcal{N}_p \left(\beta_0, \sigma^2 (\mathbf{X}^{\top} \mathbf{X})^{-1} \right).$$

Hat matrix

Let $\hat{\mathbf{Y}} = \hat{\boldsymbol{\mu}}$ be a point estimate of $\mathbb{E}(\mathbf{Y})$. Then

$$\hat{\mathbf{Y}} = \hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \mathbf{H}\mathbf{Y}$$

where $\mathbf{H} := \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is the so-called *hat matrix* (it “puts a hat” on $\mathbf{Y}$).

Properties of $\mathbf{H}$

- ▶ The *hat* (or *influence*) matrix $\mathbf{H}$ is a useful tool in model diagnosis.
- ▶ It has several interesting properties:
 - ▶ The trace of $\mathbf{H}$ is the number of identifiable parameters in the model, $\text{tr}(\mathbf{H}) = p$,
 - ▶ $\mathbf{H}$ is *idempotent*: $\mathbf{H}\mathbf{H} = \mathbf{H}$,
 - ▶ $\mathbf{H}$ is a *projection* matrix onto the space spanned by the columns of $\mathbf{X}$, and hence $\mathbf{H}\mathbf{X} = \mathbf{X}$.

Properties of $\hat{\beta}$

- ▶ Assume $\varepsilon \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ which yields $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$
 - ▶ $\mathbf{Y}$ follows a multivariate normal distribution.
- ▶ Since $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is a linear transformation of $\mathbf{Y}$ it also follows a normal distribution:

$$\hat{\beta} \sim \mathcal{N}_p(\beta_0, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}).$$

- ▶ This result is not directly useful for inference on β_0 since σ^2 is usually unknown and must be estimated.
- ▶ Let $e_i = Y_i - \mathbf{x}_i^\top \hat{\beta}$. An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n - p}.$$

- ▶ Moreover, we have

$$\frac{(n - p) \hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2.$$

Hypotheses testing

Under the assumption $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\beta_0, \sigma^2\mathbf{I}_n)$, we can answer following questions:

1. Does the data support that the j -th variable (associated with the j -th column of $\mathbf{X}$) contributes to explaining $\mathbb{E}(Y_i)$ or not?
 $\implies$ Test $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$ (Wald test).
2. Partitioning the variables as $\mathbf{X} = [\mathbf{X}_0 \ \mathbf{X}_1]$, does the data support that the variables corresponding to $\mathbf{X}_1$ *collectively* contribute to explaining $\mathbb{E}(\mathbf{Y})$ or not?
 $\implies$ Partitioning β_0 accordingly as $\beta_0 = (\beta_{0,0}^\top, \beta_{0,1}^\top)^\top$, test $H_0 : \beta_{0,1} = \mathbf{0}$ vs. $H_1 : \beta_{0,1} \neq \mathbf{0}$ (F -test).

Confidence intervals and Wald test

Distribution of the OLS estimator

Under the assumption of $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\beta_0, \sigma^2\mathbf{I}_n)$, we have

$$\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma_{\hat{\beta}_j}^2) \quad \text{with} \quad \sigma_{\hat{\beta}_j}^2 = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{j,j}.$$

Covariance of the OLS estimator

By replacing σ^2 with its unbiased estimate, we have

$$\widehat{\text{cov}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1} \quad \text{and} \quad \hat{\sigma}_{\hat{\beta}_j} = \hat{\sigma} \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{j,j}}$$

t-statistic

The above allows us to define a t-statistic which could be used to construct confidence intervals around β_j , and is the basis for hypothesis tests about individual β_j .

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0.$$

Distribution of the t-statistic

From the independence between $\hat{\beta}$ and $\hat{\sigma}^2$, we have

$$\begin{aligned} t &= \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\hat{\beta}_j}} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\hat{\sigma}^2/\sigma^2}} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\frac{\hat{\sigma}^2(n-p)}{\sigma^2(n-p)}}} \\ &\sim \frac{\mathcal{N}(0, 1)}{\sqrt{\chi_{n-p}^2/(n-p)}} \stackrel{d}{=} t_{n-p}. \end{aligned}$$

We reject H_0 at the significance level $\alpha \in (0, 1)$ if the p -value

$$\Pr(|t_{n-p}| > |t|) = 2*(1 - \text{pt}(\text{abs}(t), n-p))$$

is less than α or, equivalently, if $|t| > t_{\alpha/2, n-p}$ where $t_{\alpha/2, n-p}$ denotes the $100(1 - \alpha/2)$ -quantile of the t_{n-p} distribution.

A $100(1 - \alpha)$ confidence interval for β_j is

$$\left(\hat{\beta}_j - t_{\alpha/2, n-p} \hat{\sigma}_{\hat{\beta}_j}, \quad \hat{\beta}_j + t_{\alpha/2, n-p} \hat{\sigma}_{\hat{\beta}_j} \right)$$

Testing the Contribution of Additional Variables

Often, we want to test whether adding a **set of variables** significantly improves model fit. This commonly arises with factor variables and interaction terms, where one effect is represented by several parameters.

Assume the design matrix can be written as $\mathbf{X} = \begin{bmatrix} \mathbf{X}_0 & \mathbf{X}_1 \end{bmatrix}$ and the parameter vector can be partitioned as $\beta_0 = (\beta_{0,0}^\top, \beta_{0,1}^\top)^\top$.

We compare two **nested models**:

$$H_0 : \mathbb{E}(\mathbf{Y}) = \mathbf{X}_0\beta_{0,0} \quad \text{vs.} \quad H_1 : \mathbb{E}(\mathbf{Y}) = \mathbf{X}\beta_0.$$

An alternative way to write the hypotheses is

$$H_0 : \beta_{0,1} = 0 \quad \text{vs.} \quad H_1 : \beta_{0,1} \neq 0$$

If we reject H_0 , the additional variables significantly improve the model.

F-test for Nested Models

To test $H_0 : \beta_{0,1} = \mathbf{0}$, we construct the F -statistic

$$\begin{aligned} F &= \frac{(\|\mathbf{Y} - \mathbf{X}_0 \hat{\beta}_{0,0}\|^2 - \|\mathbf{Y} - \mathbf{X} \hat{\beta}\|^2) / \{\dim(\beta_0) - \dim(\beta_{0,0})\}}{\|\mathbf{Y} - \mathbf{X} \hat{\beta}\|^2 / \{n - \dim(\beta_0)\}} \\ &= \frac{\{RSS(\hat{\beta}_{0,0}) - RSS(\hat{\beta})\} / (p - q)}{RSS(\hat{\beta}) / (n - p)} \sim_{H_0} \frac{\chi^2_{p-q} / (p - q)}{\chi^2_{n-p} / (n - p)} =_d F_{p-q, n-p}. \end{aligned}$$

The **p-value** is computed as

$$\Pr(F_{p-q, n-p} > F) = 1 - \text{pf}(F, p-q, n-p)$$

Reject H_0 if: p-value $< \alpha$ or F exceeds the critical value.

This implies **at least one coefficient in $\beta_{0,1}$ is non-zero**.

The `lm` function in R routinely tests whether **any of the covariates are associated with Y** : $H_0 : \mathbb{E}(Y_i) = \beta_0$ vs. $H_1 : \mathbb{E}(Y_i) = \mathbf{x}_i^\top \beta_0$.

Goodness of fit R^2

- ▶ The multiple R^2 is an estimate of the proportion of the variance in the response that is explained by the regression. Given the observations $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$, we have:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{\mu}_i)^2 / n}{\sum_{i=1}^n (y_i - \bar{y})^2 / n} \quad \text{where} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

- ▶ The fraction above accounts for the *proportion* of the variance that is not explained by the regression, and therefore $0 \leq R^2 \leq 1$. The closer R^2 is to one, the more variability in y_i , $i = 1, \dots, n$ is explained by the fitted model.

Adjusted R^2

- ▶ An issue with R^2 is that it always *increases* when a new variable is added to the model, no matter how relevant that variable is for predicting Y_i .
- ▶ Part of the reason for this is that the variance estimate used to calculate R^2 , i.e. the numerator $n^{-1} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2$ under the model $\mathbb{E}(Y_i) = \mathbf{x}_i^\top \beta_0$, is biased in a way that tends to inflate R^2 . To correct this, we use the *adjusted* R^2 :

$$R_{\text{adj}}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{\mu}_i)^2 / (n - p)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)}.$$

- ▶ A large value of R_{adj}^2 indicates that the model explains the variability in the response variable well. Note that while R^2 always lies between 0 and 1, this is not necessarily the case for R_{adj}^2 .

Point Prediction

- ▶ After fitting a model using available observations $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$, one common task is to predict the expected response from the model at new values of the regressors.
- ▶ For a newly observed $\mathbf{x}_*$, a natural candidate for the predicted value of $\mu_* = \mathbb{E}(Y \mid \mathbf{x}_*)$, the mean response at $\mathbf{x}_*$ is $\hat{\mu}_* = \mathbf{x}_*^\top \hat{\boldsymbol{\beta}}$, which satisfies

$$\hat{\mu}_* \sim \mathcal{N}\left(\mathbf{x}_*^\top \boldsymbol{\beta}_0, \sigma^2 \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*\right)$$

.

Confidence vs prediction interval

- ▶ A $100(1 - \alpha)$ *confidence interval* for $\mathbb{E}(Y \mid \mathbf{x}_*)$ is designed to cover the “fixed” target $\mathbb{E}(Y \mid \mathbf{x}_*)$.

$$\hat{\mu}_* \pm t_{\alpha/2, n-p} \hat{\sigma} \sqrt{\mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*}$$

- ▶ A *prediction interval* is designed to cover a “moving” target, namely $Y_* = \mathbb{E}(Y \mid \mathbf{x}_*) + \varepsilon_*$,

$$\hat{\mu}_* \pm t_{\alpha/2, n-p} \hat{\sigma} \sqrt{1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*}.$$

- ▶ Both intervals are centered at $\hat{\mu}_*$, but the prediction interval is wider.
- ▶ Prediction intervals account for future observation Y_* variability (due to the noise ε_*), while confidence intervals only account for uncertainty in estimating the mean.

Model diagnosis

In fitting linear models, several assumptions are made, such as

- (i) the regression function is linear, i.e. for given $(Y_i, \mathbf{x}_i)$, we have $\mathbb{E}(Y_i) = \mathbf{x}_i^\top \beta_0$;
- (ii) the errors ε_i , $i = 1, \dots, n$ have zero mean and constant variance;
- (iii) the errors ε_i , $i = 1, \dots, n$ are independent;
- (iv) the errors are normally distributed, i.e. $\varepsilon_i \sim_{\text{iid}} \mathcal{N}(0, \sigma^2)$.

It is pointless to conduct any statistical inference prior to checking the plausibility of the assumptions on the data. Often, the first step in model diagnosis is to plot the (standardised) residuals against various quantities to verify these assumptions.

Raw residuals

Recall the influence (“hat”) matrix $\mathbf{H}$. The vector of raw residuals is

$$\mathbf{e} = \mathbf{Y} - \hat{\boldsymbol{\mu}} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

It follows that

$$\mathbb{E}(\mathbf{e}) = \mathbf{0}, \quad \text{cov}(\mathbf{e}) = \sigma^2(\mathbf{I}_n - \mathbf{H}).$$

With the additional normality assumption, $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$, the distribution of $\mathbf{e}$ is also normal, i.e.

$$\mathbf{e} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2(\mathbf{I}_n - \mathbf{H})).$$

Standardised residuals

Recall:

$$\mathbb{E}(e_i) = 0, \quad \text{Var}(e_i) = \sigma^2(1 - h_{ii}).$$

Estimate the standard deviation of e_i

$$\sqrt{\widehat{\text{Var}}(e_i)} = \sqrt{\hat{\sigma}^2(1 - h_{ii})}.$$

Standardised residual:

$$s_i = \frac{e_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}}.$$

Used in detecting anomalous observations or *outliers*. Observations with $|s_i| > 2$ may require further investigation.

Jackknife (externally standardised) residuals

Standardised residuals use $\hat{\sigma}$, which can be inflated by outliers, allowing outliers to escape detection.

To address this, we standardise each residual using a variance estimate computed **without that observation**.

This gives the *jackknife* (externally standardised) residual:

$$t_{\ell} = \frac{e_{\ell}}{\sqrt{\hat{\sigma}_{(\ell)}^2(1 - h_{\ell\ell})}},$$

where $\hat{\sigma}_{(\ell)}^2$ is the error variance estimated after removing observation $(Y_{\ell}, \mathbf{x}_{\ell})$, based on $n - 1 - p$ degrees of freedom.

Leave-one-out variance estimate

Let $\hat{\beta}^{(\ell)}$ be the least squares estimator computed using all observations except ℓ :

$$\hat{\beta}^{(\ell)} = \arg \min_{\beta} \sum_{i \neq \ell} (Y_i - \mathbf{x}_i^{\top} \beta)^2.$$

The corresponding variance estimate is

$$\hat{\sigma}_{(\ell)}^2 = \frac{1}{n-1-p} \sum_{i \neq \ell} \left(Y_i - \mathbf{x}_i^{\top} \hat{\beta}^{(\ell)} \right)^2.$$

Residuals e_{ℓ} and the hat matrix $\mathbf{H}$ are still computed from the model fitted using all n observations.

Checking model assumptions with residuals

Cheat sheet for checking model assumptions using residuals

Assumption	Plot	Check	Remedy
Linearity	Residuals vs $\hat{\mu}$	No pattern	Transform Y Add polynomials
Independence	Residuals vs i	No pattern	Durbin–Watson test
Equal variance	Residuals vs $\hat{\mu}$	Constant spread	Transform Y
Normality	Q–Q plot	Straight line	Less important for large n

Influential observations and outliers

- ▶ **Outliers:** large residuals (large error terms)
- ▶ **Influential observations:** unusual $\mathbf{x}_i$ values that strongly affect the fitted model

How do we measure if $\mathbf{x}_i$ is unusual?

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii})$$

As $h_{ii} \rightarrow 1$, $\text{Var}(e_i) \rightarrow 0 \implies$ the fitted value $\hat{\mu}_i$ is forced close to y_i .

Therefore, h_{ii} measures **leverage** (potential influence).

An observation is often considered **high leverage** if $h_{ii} > 2p/n$. Since $1/n \sum_{i=1}^n h_{ii} = p/n$ the cutoff $2p/n$ is **twice the average leverage**.

Cook's distance

A point with **large residual and high leverage** can strongly influence the fitted model.

Cook's distance measures how much the estimates change when observation ℓ is removed.

$$D_{\ell} = \frac{(\hat{\beta}^{(\ell)} - \hat{\beta})^{\top} \left(\widehat{\text{cov}}(\hat{\beta}) \right)^{-1} (\hat{\beta}^{(\ell)} - \hat{\beta})}{p}$$

Equivalent forms:

$$D_{\ell} = \frac{\sum_{i=1}^n (\hat{\mu}_i^{(\ell)} - \hat{\mu}_i)^2}{p\hat{\sigma}^2} = s_{\ell}^2 \frac{h_{\ell\ell}}{(1 - h_{\ell\ell})p}$$

Cook's distance combines:

- ▶ residual size (s_{ℓ}),
- ▶ leverage ($h_{\ell\ell}$).

Interpreting Cook's Distance

Rules of thumb:

- ▶ $D_\ell > 0.5 \rightarrow$ potentially influential,
- ▶ $D_\ell \geq 1 \rightarrow$ highly influential.

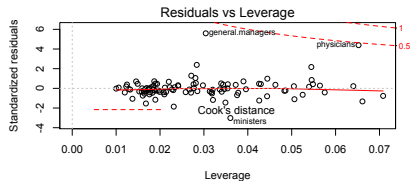
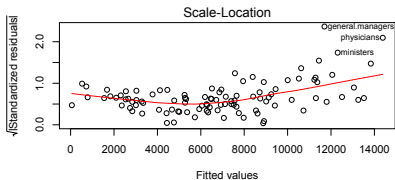
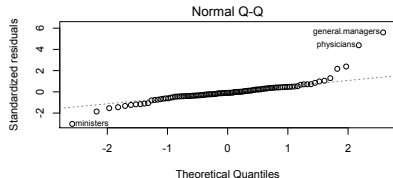
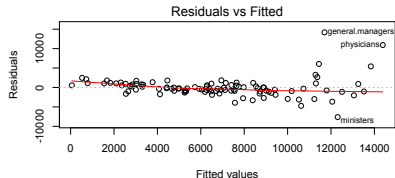
Cook's distance is related to an F statistic:

$$D_\ell \sim F_{p, n-p}$$

An influential observation is one where removing it would substantially change the parameter estimates.

Large Cook's distance $\rightarrow$ observation strongly affects the fitted model.

Residual diagnostics and influential observations – example



OLS without linearity assumption

- ▶ Let $(Y, \mathbf{X})$ be a vector of random variables.
- ▶ Suppose that true relationship is $Y = f(\mathbf{X}) + \varepsilon$ and we are interested in fitting $\mathbb{E}(Y|\mathbf{X} = \mathbf{x})$.
- ▶ The true parameter β_0 can be defined **without** the assumption of linearity **irrespective** of the functional form $f(\mathbf{X})$ through a following population-level minimization problem

$$\beta_0 = \arg \min_{\beta} \mathbb{E} \left(Y - \mathbf{X}^{\top} \beta \right)^2.$$

- ▶ The above leads to a following estimating equation $\mathbb{E}\{\mathbf{X}(Y - \mathbf{X}^{\top} \beta)\} = 0 \rightarrow \tilde{\beta} = \mathbb{E}\{(\mathbf{X}^{\top} \mathbf{X})^{-1}\} \mathbb{E}(\mathbf{X}^{\top} Y)$
- ▶ $\tilde{\beta}$ depends **only** on the joint distribution of $(Y, \mathbf{X})$.
- ▶ OLS is a sample version of the minimization problem above

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \mathbf{x}_i^{\top} \beta)^2.$$

OLS in ML – the best linear approximation

In ML the OLS is often postulated as a working model under following assumptions:

- ▶ True model: $Y = f(\mathbf{X}) + \varepsilon$
- ▶ We can write: $Y = \mathbf{X}^\top \beta + r(\mathbf{X}) + \varepsilon$ where $r(\mathbf{X}) = f(\mathbf{X}) - \mathbf{X}^\top \beta$.
- ▶ Orthogonality condition to derive β :
$$\mathbb{E}\{\mathbf{X}(Y - \mathbf{X}^\top \beta)\} = \mathbb{E}[\mathbf{X}\{r(\mathbf{X}) + \varepsilon\}] = 0$$
- ▶ If we further assume exogeneity of random noise $\mathbb{E}(\varepsilon \mid \mathbf{X}) = 0$ this leads to orthogonality of $r(\mathbf{X})$ by
$$\begin{aligned}\mathbb{E}[\mathbf{X}\{r(\mathbf{X}) + \varepsilon\}] &= \mathbb{E}\{\mathbf{X}r(\mathbf{X})\} + \mathbb{E}(\mathbf{X}\varepsilon) \\ &= \mathbb{E}\{\mathbf{X}r(\mathbf{X})\} + \mathbb{E}\{\mathbf{X}\mathbb{E}(\varepsilon \mid \mathbf{X})\} \\ &= \mathbb{E}\{\mathbf{X}r(\mathbf{X})\} + \mathbb{E}(\mathbf{X}0) = \mathbb{E}\{\mathbf{X}r(\mathbf{X})\} = 0\end{aligned}$$

Why the orthogonality decomposition is important

- 1. Projection interpretation (geometry).** The linear predictor $\mathbf{X}^\top \beta$ is the orthogonal projection of Y onto the span of X . Linear regression therefore decomposes Y into a linear component and an orthogonal residual $r(\mathbf{X}) + \varepsilon$.
- 2. Interpretation of the coefficients.** Because the nonlinear component $r(\mathbf{X})$ is orthogonal to $\mathbf{X}$, the vector β captures all of the linear relationship between $\mathbf{X}$ and Y . In other words, β represents the coefficients of the best linear approximation to $\mathbb{E}(Y \mid X)$, even when **the true relationship is nonlinear**.
- 3. Moment condition and estimation robustness.** The condition $\mathbb{E} \left[\mathbf{X} \left(Y - \mathbf{X}^\top \beta \right) \right] = 0$ defines β through moment equations/orthogonality condition rather than through a correct model specification. This is the basis of M-estimation or GMM, and explains why linear regression estimators remain meaningful and consistent for the best linear predictor even when the linear model is misspecified.

Linear models

- ▶ Linear models
 - ▶ Linear regression
 - ▶ Logistic regression
 - ▶ Linear discriminant analysis

Setup of Logistic Regression

- ▶ Random variables $Y_i \in \{0, 1\}$, $\mathbf{X}_i \in \mathbf{R}^p$
- ▶ Data/realizations $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$
- ▶ Y_i independent conditional on $\mathbf{X}_i$

Goal: Find a model for a conditional probability

$$\pi_i = \mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = \mathbb{E}(Y_i \mid \mathbf{x}_i).$$

Why not linear regression? $Y_i = \pi_i + \varepsilon_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

- ▶ The linear model does not return a binary quantity.
- ▶ Even relaxing the normality assumption, the errors ε_i cannot be homoscedastic, as for a Bernoulli Y_i ,
 $\text{Var}(Y_i = 1 \mid \mathbf{x}_i) = \pi_i(1 - \pi_i)$.
- ▶ The linear predictor $\mathbf{x}_i^\top \boldsymbol{\beta}$ allows values $\pi_i < 0$ or $\pi_i > 1$, which are impossible for probabilities.

Logistic Regression

Idea

Link $\mathbb{E}(Y_i) = \pi_i$ and the linear predictor $\mathbf{x}_i^\top \boldsymbol{\beta}$ with a *link function*.

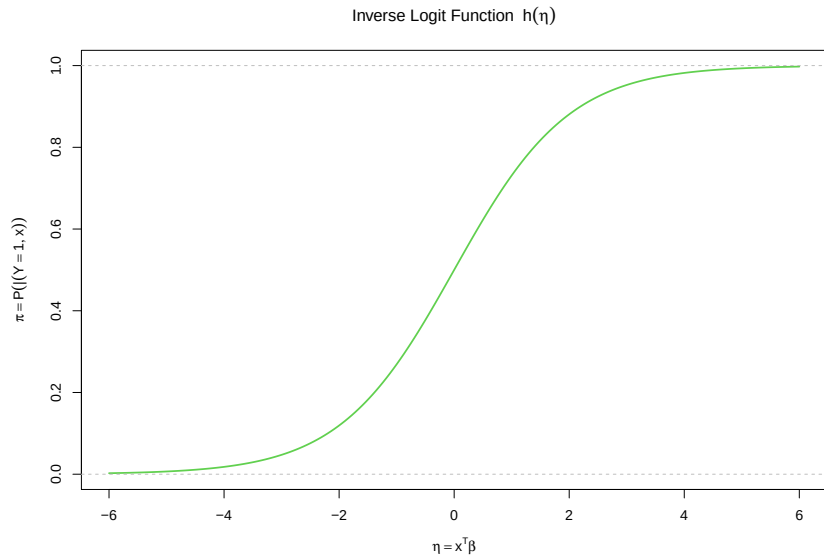
- ▶ Logit link function $g(\pi_i) = \log \frac{\pi_i}{1-\pi_i}$
- ▶ Response (inverse logit) function $h : \mathbb{R} \rightarrow (0, 1)$

$$\pi_i = P(Y_i = 1 | \mathbf{x}_i) = h(\mathbf{x}_i^\top \boldsymbol{\beta}) = h(\eta_i)$$

$$h(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} = \frac{1}{1 + \exp(-\eta_i)}$$

$$g(\pi_i) = h^{-1}(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Plot with inverse logit



Odds and log-odds

- ▶ logarithmic odds, log-odds: $\log\left(\frac{\pi_i}{1-\pi_i}\right)$
- ▶ linear predictor: $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The choice of inverse logit h as a response function yields a *multiplicative model for the odds*

$$\frac{\pi_i}{1 - \pi_i} = \frac{\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = 0 \mid \mathbf{x}_i)} = \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdots \exp(\beta_p x_{ip}).$$

Odds and log-odds vs. increase of x_{i1}

A unit increase of x_{i1} increases the odds by a multiplicative factor $\exp(\beta_1)$

$$\begin{aligned}\frac{\mathbb{P}(Y_i = 1 \mid x_{i1} + 1, \dots)}{\mathbb{P}(Y_i = 0 \mid x_{i1} + 1, \dots)} &= \exp(\beta_0) \exp(\beta_1(x_{i1} + 1)) \cdots \exp(\beta_p x_{ip}) \\ &= \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdots \exp(\beta_p x_{ip}) \exp(\beta_1) \\ &= \frac{\mathbb{P}(Y_i = 1 \mid x_{i1}, \dots)}{\mathbb{P}(Y_i = 0 \mid x_{i1}, \dots)} \exp(\beta_1).\end{aligned}$$

At the same time, a unit increase of x_{i1} increases the log-odds by the additive constant β_1 :

$$\log \frac{\mathbb{P}(Y_i = 1 \mid x_{i1} + 1, \dots)}{\mathbb{P}(Y_i = 0 \mid x_{i1} + 1, \dots)} = \log \frac{\mathbb{P}(Y_i = 1 \mid x_{i1}, \dots)}{\mathbb{P}(Y_i = 0 \mid x_{i1}, \dots)} + \beta_1.$$

Interpretation: odds and log-odds

The logistic regression model assumes a linear relationship between the log-odds of Y and the independent variables $\mathbf{X}$.

Interpretation

- ▶ When x_k increases by one unit, the logarithmic odds of Y change by β_k (increase if $\beta_k > 0$, decrease if $\beta_k < 0$, stay constant if $\beta_k = 0$), c.p.
- ▶ When x_k increases by one unit, the odds of Y change by the factor $\exp(\beta_k)$, c.p.
- ▶ The odds ratio of Y at $x_k + 1$ and Y at x_k is $\exp(\beta_k)$, c.p.

Probability, Odds, and Log-odds

Probability	0.01	0.05	0.5	0.95	0.99
Odds	1/99	1/19	1	19	99
Log-odds	-4.6	-2.9	0	2.9	4.6

ML estimation

- ▶ Input $\mathbf{X}_{n \times p}$ and $\mathbf{Y}_{n \times 1} \in \{0, 1\}^n$
- ▶ Bernoulli distribution $f(y_i; \pi_i) = \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$, $y_i \in \{0, 1\}$
- ▶ Recall that $\pi_i = h(\eta_i) = h(\mathbf{x}_i^\top \boldsymbol{\beta})$
- ▶ Maximum Likelihood Estimation (MLE) problem

$$\begin{aligned} L(\boldsymbol{\beta}; \mathbf{Y} = \mathbf{y}) &= \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\ &= [h(\mathbf{x}_i^\top \boldsymbol{\beta})^{y_i} (1 - h(\mathbf{x}_i^\top \boldsymbol{\beta}))^{1-y_i}] \\ \log L(\boldsymbol{\beta}; \mathbf{y}) &= \sum_{i=1}^n [y_i \log\{h(\mathbf{x}_i^\top \boldsymbol{\beta})\} + (1 - y_i) \log(1 - h(\mathbf{x}_i^\top \boldsymbol{\beta}))] \end{aligned}$$

Estimating parameters in logistic regression

- ▶ ML estimator defined as

$$\hat{\beta}_{ML} = \arg \max L(\beta) = \arg \max (\ln L(\beta))$$

- ▶ Taking the derivative with respect to (w.r.t.) β , equating it to 0 and using $h' = h(1 - h)$ results in the score equation for $\hat{\beta}_{ML}$:

$$s(\hat{\beta}_{ML}) = \sum_{i=1}^n \left(Y_i - h(\mathbf{x}_i^\top \hat{\beta}_{ML}) \right) \mathbf{x}_i = 0.$$

Cross-entropy in ML

- ▶ Cross-entropy loss: $CE = -\log L(\beta; \mathbf{y})$ (v. popular in ML)
- ▶ *Maximizing the log-likelihood $\leftrightarrow$ minimizing cross-entropy*, i.e., minimizing the difference between true outcomes y_i and predicted probabilities π_i .

Asymptotic distribution of the MLE

Theorem

Let $(\mathbf{Z}_i)_{i=1}^n$ be independent random p -vectors satisfying $\mathbb{E}(\mathbf{Z}_i) = \mathbf{0}$ and $\mathbb{E}(\mathbf{Z}_i \mathbf{Z}_i^\top) = \boldsymbol{\Sigma}_i$, and let $\boldsymbol{\Sigma} = \frac{1}{n} \sum_{i=1}^n \boldsymbol{\Sigma}_i$. Then, under [Lindeberg condition](#) on the tail of $\mathbf{Z}_i$, i.e., for every $\varepsilon > 0$,

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E} \left[\|\mathbf{Z}_i\|^2 \mathbf{1}_{\{\|\mathbf{Z}_i\| > \varepsilon \sqrt{n}\}} \right] \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

we have a following result

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{as } n \rightarrow \infty,$$

where $\xrightarrow{d}$ denotes convergence in distribution.

Asymptotic distribution of the MLE

Corollary 1

Consider independent random variables $Y_1, \dots, Y_n$ from some distribution F_{β_0} with β_0 denoting the true parameter value. Also denote the associated log-likelihood function by $\ell(\beta) = \sum_{i=1}^n \ell_i(\beta)$.

As $n \rightarrow \infty$, $\frac{1}{\sqrt{n}} \frac{\partial \ell}{\partial \beta}(\beta_0) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, I_1(\beta_0))$, where

$$I_1(\beta) = \left[-\frac{1}{n} \mathbb{E} \left(\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k}(\beta) \right) \right]_{j,k=1}^p = \frac{1}{n} I(\beta).$$

is the scaled *Fisher information matrix*

Corollary 2

Let $\hat{\beta}$ be the maximum likelihood estimator of β_0 , which satisfies $\frac{\partial \ell}{\partial \beta}(\hat{\beta}) = \mathbf{0}$. By Corollary 1, as $n \rightarrow \infty$, it follows that,

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, (I_1(\beta_0))^{-1})$$

Asymptotic confidence intervals and hypotheses testing

- ▶ Estimate the covariance $\text{cov}(\hat{\beta})$ of the MLE by

$$\widehat{\text{cov}}(\hat{\beta}) = (nl_1(\hat{\beta}))^{-1}.$$

- ▶ The standard error of $\hat{\beta}_j$ is the square root of the j -th diagonal element:

$$\widehat{\text{s.e.}}(\hat{\beta}_j) = \sqrt{\left[(l(\hat{\beta}))^{-1}\right]_{j,j}}.$$

- ▶ By asymptotic normality of the MLE, an approximate $100(1 - \alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm z_{\alpha/2} \widehat{\text{s.e.}}(\hat{\beta}_j).$$

- ▶ If this interval contains 0, we do not reject $H_0 : \beta_j = 0$ at the significance level $\alpha = 0.05$.

Example: Logistic Regression with Iris data

```
suppressWarnings(suppressMessages(library(tidyverse)))  
data("iris")  
head(iris)
```

##	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
## 1	5.1	3.5	1.4	0.2	setosa
## 2	4.9	3.0	1.4	0.2	setosa
## 3	4.7	3.2	1.3	0.2	setosa
## 4	4.6	3.1	1.5	0.2	setosa
## 5	5.0	3.6	1.4	0.2	setosa
## 6	5.4	3.9	1.7	0.4	setosa

```
iris2 <- iris%>%filter(Species%in% c("versicolor", "virginica"))  
model <- glm(Species ~ Petal.Length + Petal.Width, data = iris2,  
             family = "binomial")
```

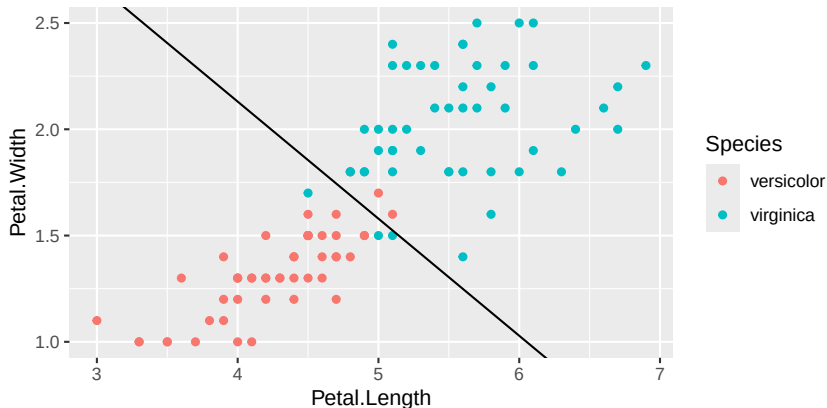
Example: Logistic Regression with Iris data

```
summary(model)
```

```
##
## Call:
## glm(formula = Species ~ Petal.Length + Petal.Width, family = "binomial",
##      data = iris2)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -45.272     13.610  -3.327 0.000879 ***
## Petal.Length    5.755      2.306   2.496 0.012565 *
## Petal.Width    10.447      3.755   2.782 0.005405 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 138.629  on 99  degrees of freedom
## Residual deviance:  20.564  on 97  degrees of freedom
## AIC: 26.564
##
## Number of Fisher Scoring iterations: 8
```

Example: Logistic Regression with Iris data

```
ggplot(iris2) +  
  geom_point(aes(x = Petal.Length, y = Petal.Width, color = Species)) +  
  geom_abline(  
    slope = -model$coefficients[2] / model$coefficients[3],  
    intercept = -model$coefficients[1] / model$coefficients[3]  
  )
```



Linear models

- ▶ Linear models
 - ▶ Linear regression
 - ▶ Logistic regression
 - ▶ **Linear discriminant analysis**

Linear Discriminant Analysis (LDA)

Idea: we model the distribution of X in each class and then use Bayes' rule to compute class probabilities (**generative classification method**)

- ▶ Each class has a class-conditional density
 $f_k(x) = f_{X|Y}(x|Y = k)$
- ▶ Prior probability $P(Y = k) = \pi_k$
- ▶ Classification is based on posterior probabilities
- ▶ Bayes Rule (Reminder): for events A and B ,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

Then, using Bayes' rule:

$$P(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^K \pi_{\ell} f_{\ell}(x)}$$

Classify x to the class with the largest posterior probability.

Gaussian Class-Conditional Model

Assume that in each class

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \sigma^2).$$

Then the class-conditional density is

$$f_k(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)$$

Posterior probability:

$$p_k(x) = P(Y = k \mid X = x) = \frac{\frac{\pi_k}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{\ell=1}^K \frac{\pi_\ell}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_\ell)^2\right)}$$

Univariate case: $K = 2$

If $K = 2$ and $\pi_1 = \pi_2$, we assign an observation to class 1 if:
 $\log \frac{p_1(x)}{p_2(x)} > 0 \rightarrow \log(p_1(x)) > \log(p_2(x))$. Then

$$\begin{aligned} & \log(p_1(x)) - \log(p_2(x)) \\ &= \log(\pi_1) - \log(\pi_2) - \frac{(x - \mu_1)^2}{2\sigma^2} + \frac{(x - \mu_2)^2}{2\sigma^2} \\ &= \log(\pi_1) - \log(\pi_2) - \frac{x^2 - 2x\mu_1 + \mu_1^2}{2\sigma^2} + \frac{x^2 - 2x\mu_2 + \mu_2^2}{2\sigma^2} \\ &= \log(\pi_1) - \log(\pi_2) + \frac{x(\mu_1 - \mu_2)}{\sigma^2} - \frac{\mu_1^2 - \mu_2^2}{2\sigma^2} > 0 \end{aligned}$$

If $\pi_1 = \pi_2$, then the decision boundary satisfies

$$x = \frac{\mu_1 + \mu_2}{2}.$$

LDA - Iris data

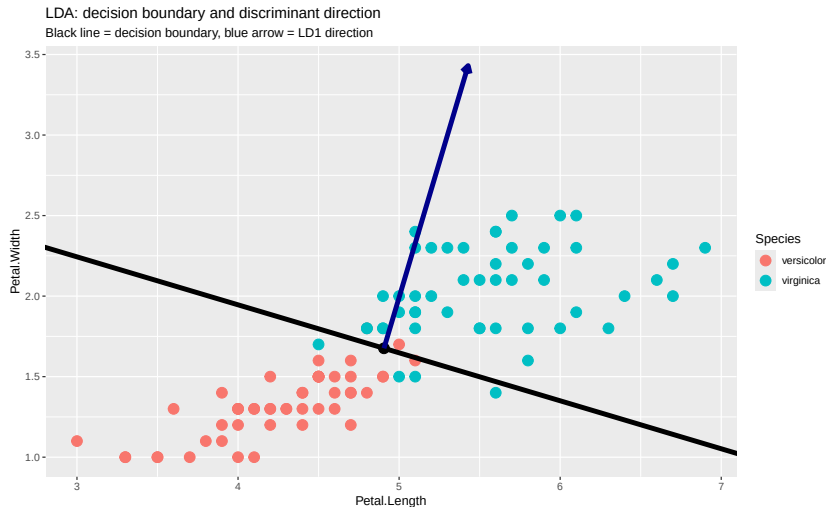
```
suppressWarnings(suppressMessages(library(MASS)))

model <- suppressWarnings(lda(Species ~ Petal.Length + Petal.Width,
                             data = iris2))

model

## Call:
## lda(Species ~ Petal.Length + Petal.Width, data = iris2)
##
## Prior probabilities of groups:
## versicolor  virginica
##          0.5          0.5
##
## Group means:
##          Petal.Length Petal.Width
## versicolor          4.260          1.326
## virginica           5.552          2.026
##
## Coefficients of linear discriminants:
##              LD1
## Petal.Length 0.8712821
## Petal.Width  2.9247035
```

LDA Iris data – decision boundary and discriminant direction



Recovering the LDA decision boundary

Let $w = \text{model}\$scaling[,1]$ be the **orientation vector** returned by `lda()`. It determines how the classes are separated.

The decision boundary is the line **perpendicular** to w and passing through the midpoint of the class means,

$$\text{mid} = \frac{\mu_1 + \mu_2}{2}.$$

If $w = (w_1, w_2)^\top$, then the boundary has equation

$$x_2 = -\frac{w_1}{w_2}x_1 + \left(\text{mid}_2 + \frac{w_1}{w_2}\text{mid}_1\right).$$

Equivalently, a line perpendicular to w which passes through mid can be found as follows:

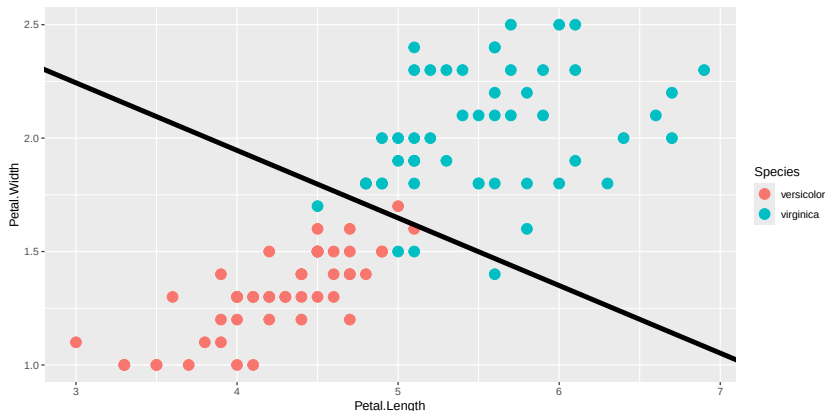
$$w_1(x_1 - \text{mid}_1) + w_2(x_2 - \text{mid}_2) = 0.$$

Recovering the LDA decision boundary – Iris data

```
w <- model$scaling[,1]    # determines the orientation  
# of the boundary  
means <- model$means      # estimated class means  
mid <- colMeans(means)    # midpoint between class means  
  
# slope and intercept of decision boundary  
slope <- -w[1] / w[2]  
intercept <- mid[2] - slope * mid[1]
```

LDA Iris data – decision boundary

```
ggplot(iris2, aes(Petal.Length, Petal.Width,  
  color = Species)) + geom_point(size = 4) +  
  geom_abline(slope = slope, intercept = intercept,  
    linewidth = 2)
```



Fisher's Linear discriminant analysis

Idea

Choose the **optimal** projection vector w , which determines the orientation of the LDA decision boundary, so that in the projected lower-dimensional space the classes are as well separated as possible.

Illustration $K = 2$

- ▶ Projection of data point: $z_i = w^\top x_i$, where w is a 2×1 vector and z_i is a scalar.
- ▶ Projection of mean: $m_i = w^\top \mu_i$ $i = 1, 2$
- ▶ Between-class scatter matrix: $S_B = (m_1 - m_2)^2$
- ▶ Within-class scatter matrix:
 - ▶ $S_{in} = \sum_{z_i \in \text{class}\{1\}} (z_i - m_1)^2 + \sum_{z_i \in \text{class}\{2\}} (z_i - m_2)^2$
- ▶ Optimization: choose w to maximize

$$\operatorname{argmax}_w \frac{S_B}{S_{in}} = \operatorname{argmax}_w \frac{w^\top (\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top w}{w^\top (\Sigma_1 + \Sigma_2) w}.$$

Linear discriminant analysis

- ▶ Further topics:
 - ▶ Estimation of w in Fisher LDA
 - ▶ Quadratic discriminant analysis
 - ▶ $p > n$
- ▶ Comparison between Logistic regression and LDA:
 - ▶ Both Logistic Regression and LDA produce **linear boundaries**
- ▶ LDA assumes that the observations are drawn from the normal distribution with common variance in each class, while logistic regression does not have this assumption.
- ▶ LDA would do better than Logistic Regression if the assumption of normality holds, otherwise logistic regression may outperform LDA.

Quick summary about linear models

- ▶ Advantages:
 - ▶ Interpretable
 - ▶ Relatively easy to fit
 - ▶ Well-established software packages
- ▶ Disadvantages:
 - ▶ Strong distribution assumptions
 - ▶ Not flexible enough

References

- ▶ [ESLII](#): Chapters 3.1, 4.1-4.4
- ▶ [PRML](#) Chapters 3.1, 4.1
- ▶ [PML](#): Chapters 9-12